Roll No. | | | | | | | |

ODD END TERM EXAMINATION NOVEMBER/DECEMBER - 2025

Name of the Course: **B.Tech., ME** Semester: V[th]

Name of the Paper: **Mechanical Measurements and Metrology** Course Code: **(TME 510)**

Time: 3 Hours Maximum Marks: 100

Note:-

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q1	**(20 marks)**	
(a)	Explain the construction, classification, and uses of Indian Slip Gauge Sets (M-87 and M-112) as per IS standards.	**CO5 and CO1**
(b)	Differentiate between line standards and end standards with suitable examples. Discuss their advantages and limitations.	
(c)	During calibration of an end bar of 250 mm, an optical comparator shows a deviation of +0.008 mm. Determine the actual length and percentage error.	
Q2	**(20 marks)**	
(a)	What is Taylor's Principle in gauge design? Illustrate with a neat sketch and explain its application in GO and NO-GO gauges.	**CO2 and CO5**
(b)	Explain hole basis and shaft basis systems of fits with diagrams. State their advantages and applications.	
(c)	For a shaft of nominal size 40 mm, the tolerance on hole is H7 (+0.025/0.000 mm) and on shaft is f7 (−0.025/−0.050 mm). Find the type of fit and maximum and minimum clearances.	
Q3	**(20 marks)**	
(a)	Describe the construction and working of LVDT. Explain how it is used as a displacement measuring device.	**CO3 and CO5**
(b)	Discuss the principle of interferometer and explain how it is used for flatness measurement using optical flats.	
(c)	In a three-wire method, the measured diameter over wires is 25.683 mm, wire diameter is 0.5 mm, and pitch is 1.5 mm. Calculate the effective diameter of the screw thread.	
Q4	**(20 marks)**	
(a)	Discuss different types of errors in measurement and explain the methods to minimize systematic and random errors	**CO4 and CO5**
(b)	A measuring instrument gives five readings of a 50.00 mm gauge block as 49.98, 50.01, 49.99, 50.02, and 49.97 mm. Calculate mean, deviation, and accuracy if the true value is 50.00 mm.	
(c)	Write short notes on form and finish measurement techniques. Explain how surface roughness is measured.	
Q5	**(20 marks)**	
(a)	Explain the principle, construction, and working of a thermocouple and an optical pyrometer for temperature measurement.	**CO5 and CO6**
(b)	What is a strain gauge? Explain the gauge factor and the procedure for mounting strain gauges. Discuss various methods of strain measurement.	
(c)	Compare the Bridgman, McLeod, and Pirani gauges used for pressure measurement. Mention their ranges and limitations.	